ASSOCIATED DETAIL SPECIFICATION

RESISTORS, FIXED, FILM, HIGH VOLTAGE, STYLES RHV36, RHV37, RHV38, RHV39, RHV40, RHV41, RHV42, RHV43, RHV44, RHV45, RHV46, RHV47

This specification is approved for use within the National Aeronautics and Space Administration, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RHV36, RHV37, RHV38, RHV39, RHV40, RHV41, RHV42, RHV43, RHV44, RHV45, RHV46, and RHV47 resistors.

1.2 Military part number. Resistors covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:



The coded number shall be derived in accordance with MIL-R-49462.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-39032 - Resistors, Packaging of.

MIL-R-49462 - Resistors, Fixed, Film, High Voltage, General Specification for.

(Copies of the specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA Goddard Space Flight Center, Code 311.A, Greenbelt, MD 20771, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-49462, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be specified in table I, based on a full load operation at an ambient temperature of 125°C. For operation at temperatures higher than 125°C, derate in accordance with figure 2.

TABLE I. Power rating.

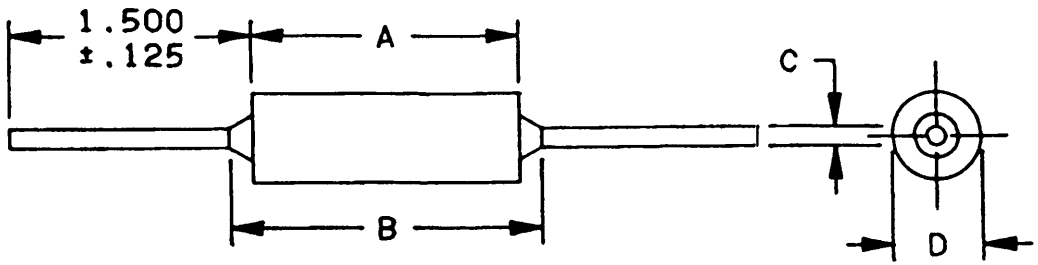
Resistor style	Power rating (watts)	Style
RHV36	0.5	A
RHV37	1.0	B
RHV38	1.5	C
RHV39	2.0	D
RHV40	2.5	E
RHV41	3.0	F
RHV42	5.0	G
RHV43	6.0	H
RHV44	7.5	J
RHV45	8.0	K
RHV46	10.0	M
RHV47	15.0	N

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table II.

TABLE II. Minimum and maximum nominal resistance values.

Resistor style	Resistance value (ohms)		Style
	Minimum	Maximum	
RHV36	1k	1G	A
RHV37	1.2k	2.49G	B
RHV38	1.7k	10G	C
RHV39	2k	10G	D
RHV40	1.5k	7.5G	E
RHV41	2k	10G	F
RHV42	3k	15G	G
RHV43	4k	20G	H
RHV44	600	750M	J
RHV45	800	1000M	K
RHV46	1k	1270M	M
RHV47	1k	2000M	N

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Inches	mm
.002	0.05
.032	0.81
.125	3.18
1.500	38.10

	Dimensions							
	RHV36	RHV37	RHV38	RHV39	RHV40	RHV41	RHV42	RHV43
A	0.400 (10.2)	0.775 (19.7)	1.275 (32.4)	1.525 (38.7)	1.550 (39.4)	2.050 (52.1)	3.050 (77.5)	4.050 (102.9)
B max	0.470 (11.9)	0.845 (21.5)	1.345 (34.2)	1.595 (40.5)	1.634 (41.5)	2.134 (54.2)	3.134 (79.6)	4.134 (105.0)
C	0.25 ±.002 (0.64)	0.25 ±.002 (0.64)	0.25 ±.002 (0.64)	0.25 ±.002 (0.64)	.032 ±.002 (.081)	.032 ±.002 (.081)	.032 ±.002 (.081)	.032 ±.002 (.081)
D	.135 ±.015 (3.43 ±0.38)	.135 ±.015 (3.43 ±0.38)	.135 ±.015 (3.43 ±0.38)	.135 ±.015 (3.43 ±0.38)	.270 ±.015 (6.86 ±0.38)	.270 ±.015 (6.86 ±0.38)	.270 ±.015 (6.86 ±0.38)	.270 ±.015 (6.86 ±0.38)

NOTES:
1. Dimensions are given in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.

FIGURE 1. Resistor, fixed, film, high voltage.

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	RHV44	RHV45	RHV46	RHV47
A	3.125 ±.060 (79.38 ±1.52)	4.000 ±.120 (101.60 ±3.05)	5.000 ±.120 (127.00 ±3.05)	6.000 ±.120 (152.00 ±3.05)
B max	3.265 (82.93)	4.200 (106.68)	5.200 (132.08)	6.200 (157.48)
C	.040 ±.002 (1.02 ±0.05)	.040 ±.002 (1.02 ±0.05)	.040 ±.002 (1.02 ±0.05)	.040 ±.002 (1.02 ±0.05)
D	.315 ±.040 (8.00 ±0.76)	.315 ±.040 (8.00 ±0.76)	.315 ±.040 (8.00 ±0.76)	.350 ±.040 (8.89 ±1.02)

FIGURE 1. Resistors, fixed film, high voltage - Continued.

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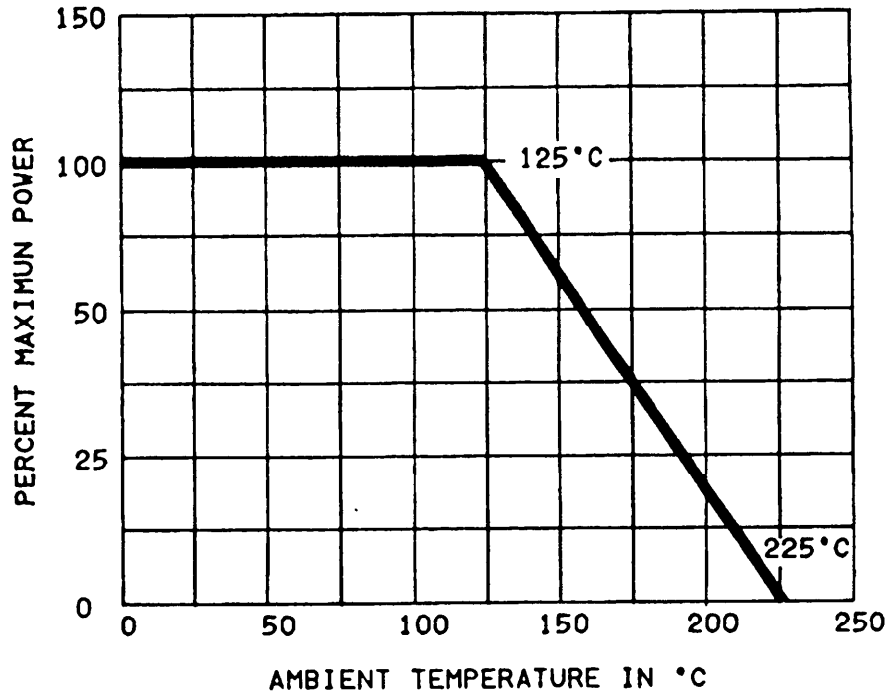


FIGURE 2. Derating curve.

3.5 Maximum voltages. Maximum voltages shall be in accordance with table III.

TABLE III. Maximum voltages.

Resistors style	Voltage (volts maximum) (kV)	Style
RHV36	1.5	A
RHV37	2.0	B
RHV38	5.0	C
RHV39	6.0	D
RHV40	11.0	E
RHV41	15.0	F
RHV42	22.5	G
RHV43	30.0	H
RHV44	15.0	J
RHV45	20.0	K
RHV46	25.0	M
RHV47	30.0	N

3.6 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance.

Symbol	Resistive tolerance percent (±)
F	1.0
G	2.0
J	5.0

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3.7 Outgassing. The outgassing requirement shall be in accordance with MIL-R-49462.

3.8 Resistor core. Solid glass or solid ceramic cores shall be utilized in the construction of these resistors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-49462.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

6.1 Intended use. Resistors are intended for use in electronic circuits where high voltages and high resistances are required.

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5905-K287)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-49462/4(NASA)	RESISTORS, FIXED, FILM, HIGH VOLTAGE, STYLES RHV36 through RHV47

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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